

CAMPUS PRODUCTIVITY TOOL: Student Portal Management System

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Abstract— Students and Educational Institutes encounter various challenges because of the existence of independent systems in managing academics, preparation of study material, conducting examinations, and handling campus placements. These independent procedures result in increased manual efforts, reduced accessibility to academic resources, difficult evaluation of results and performance, and ineffective management of placement procedures, etc. This paper proposes a Campus Productivity Tool comprising an independent set of role-based web applications to handle all academic, examination, AI-based learning, and placement activities in one place. The proposed system offers role-based management of students and teachers' profile, timetable, attendance, examination material, company's profile, eligibility details and interview schedules in one place.

The system also comprises an AI-based document processor to extract relevant information, summary and Multiple Choice Questions (MCQs) from the uploaded study material by implementing a Retrieval-Augmented Generation (RAG) model for effective document summarization. The examination module enables automated marking by comparing the students' answers with standard answers and calculating similarity scores. Similarly, the placement module handles eligibility criteria, interview schedules and provides an option to share interview schedules and reminder emails.

By emphasizing role-based features, document retrieval and processing, Large Language Models, automated marking and management of student and teacher's profiles, this paper proposes an efficient campus productivity tool to provide students access to academic resources and placement activities effortlessly.

Keywords—Campus Productivity Tool; Student Portal Management System; Retrieval-Augmented Generation; RAG; Large Language Models; Document Summarization; MCQ Generation; Automated Answer Evaluation; Role-Based Access Control; Student Management; Campus Placement; Resume Eligibility; AI-Based Education.

I. INTRODUCTION

Email has become the primary way of communication. Educational organizations tend to perform a wide range of educational and student-related activities including management of student and teacher data and timetables, attendance, exams, study materials and placement

activities. Many organizations have various systems or partial manual processing where students, teachers and administrators are required to access and manage information from different sources. Not only does it contribute to the administrative burden but is challenging to handle to get the accurate information about students' academic information. The large amount of study materials that have been digitized is a problem for students retrieving important information, preparing for exams and generating relevant learning resources. Conventional student management systems usually include tools for profile management, attendance management, timetable management, examination activities and placement information. However, they do not provide intelligent processing and automatic assistance such as extracting valuable information from educational documents or aiding in autonomous learning.

Students not only have to read through the long study materials to extract relevant concepts and create questions for self-testing but also examination answer evaluation and placement eligibility checking require manual efforts. The recent advancements on Natural Language Processing (NLP), Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) can be leveraged to achieve automations in various educational activities. RAG-based methods can be used to extract relevant information from given knowledge sources and provide to the language model for generating responses that are grounded on the provided documents. Educational research has seen the effectiveness of using RAG-based methods for context-aware student assistance and personalized learning alike. Similarly, LLM-based

approaches have been used for automatically generating Multiple Choice Questions (MCQs) from educational content thus eliminating the need for manual question creation. To address the aforementioned limitations, this paper proposes a Campus Productivity Tool, a unified role-based web application that integrates academic management, AI-assisted document, examination and campus placement activities. The tool provides distinct functionalities for administrators, teachers and students. Administrators can manage the student and teacher profiles, company profiles, eligibility requirements, correct answers and interview schedules. The teachers can manage the timetable, attendance, question papers and answer information whereas the students can view their academic information and upload study materials and resumes.

The proposed system utilizes a Retrieval-Augmented Generation (RAG) pipeline in processing the uploaded study materials. The documents are collected, preprocessed, chunked appropriately and embeddings generated and stored in a vector database. When a student requests for information or learning content, relevant document segments are retrieved based on their semantic similarities and used as context. Based on the retrieved information, the system can automatically generate meaningful summaries and Multiple Choice Questions (MCQs) on the students' uploaded learning materials. This way, students can get relevant information from lengthy documents while also preparing for their examinations using the automatically generated MCQs. Similar studies have been made on RAG-based educational systems and automatic MCQ generation from documents demonstrating the effectiveness of combining document retrieval and LLM-based question generation.

The system also includes an examination and answer evaluation module where students' answers can be compared against correct answers using similarity-based evaluation. The campus placement module handles company profile, eligibility requirements and interview schedules. Students' academic performance can be compared with the company's eligibility requirements and interview-related notifications sent via email. The proposed system aims to minimize manual efforts, provide accessible educational resources and centralize academic and student-related activities into a unified platform, offering a convenient environment for students and staff.

II. LITERATURE REVIEW

Artificial Intelligence (AI) has been increasingly utilized in education to cater to individualized learning, intelligent tutoring, automated assessments, and student academic aid. Recent studies have explored the application of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and machine learning techniques to enhance the accessibility and utility of educational resources. Hang et al. [1] presented MCQGen, an LLM-based framework that utilizes RAG and prompt engineering for automated multiple-choice question generation. The framework relies on external knowledge and self-refinement techniques to generate contextually relevant questions, demonstrating the potential of generative AI for personalized learning. Similarly, studies have investigated the application of RAG-based

approaches for reliable and context-aware academic assistance. A RAG-based chatbot solution developed by the authors of [2] utilized a course guide as the primary knowledge source to provide personalized educational guidance while minimizing the possibility of unsupported responses. Additionally, the study [3] presented a web-based intelligent tutoring system that utilized an LLM and RAG mechanism for interactive question-answering in programming education. The study demonstrated the applicability of RAG for domain-specific student assistance with the implementation of LangChain and a history-aware retriever. The application of RAG to personal learning has been further investigated by studies exploring modular system designs. OpenRAG [4] presented an open-source personalized learning architecture, comprised of a generator, user interface, indexing, retrieval, and orchestration components. Similar to the prior studies, the architecture emphasized on modularity, content accuracy, computational efficiency, and adaptability in education. An RAG-based AI assistant that was driven by Llama-3, LangChain, Chroma, and document-processing technologies was proposed in [5] to provide personalized university support from PDF, image, and Word documents. Such studies demonstrated the synergistic use of document retrieval and generative models to provide centralized academic assistance.

Automated assessments and question generation have also been extensively explored. The PAMGen framework [6] presented an approach that integrates LLMs, RAG, and prompt engineering techniques such as Chain-of-Thought and self-refinement, to generate MCQs with varying levels of difficulty and contextual relevance. Another recent study [7] investigated AI-assisted MCQ generation for engineering education, by extracting learning outcomes from lecture materials and automating the question and explanation generation process with the utilization of multimodal language models.

Thus, a number of studies have demonstrated that generative AI can significantly reduce the manual effort involved in preparing learning and assessment-related materials, while human evaluation is still crucial in order to ensure the quality of the generated questions. Machine learning has also found application in student placement and career-related prediction. The Student Placement Chance Prediction Model [8] employed XGBoost and student attributes, such as academic results, technical skills, training experiences, and projects, to predict placement outcomes and organization categories. Another study [9] investigated probabilistic student placement assessment with the utilization of machine learning, and analyzed the relationship between emotional quotient and placement outcomes. Thus, the studies demonstrate the potential of data-driven approaches for supporting student placement activities, as well as gaining important insights relevant to the domain of employment outcomes.

Recent research has emphasized the significance of reliability and evaluation in educational generative AI systems. GenEdu [10] utilizes RAG and adaptive quiz generation, level-aware explanations, document-grounded analysis, and a dual-layer evaluation mechanism. Its evaluation results demonstrate improvements in semantic alignment and faithfulness, while minimizing hallucination rates, when RAG is incorporated. Thus, such findings emphasize the significance of grounding the generated educational content on relevant source

documents, rather than relying on the internal knowledge of an LLM.

While the aforementioned studies demonstrate significant advancements regarding AI-assisted learning, tutoring systems, automated question generation, and student-related placement prediction, many approaches focus on individual facets of education, without integrating academic management, document processing, examination-related support, and campus placement within a unified student-oriented platform. The proposed Campus Productivity Tool addresses this research gap, since it integrates role-based student management, AI-based document processing and RAG-driven summarization and MCQ generation, similarity-based answer evaluation, company eligibility management, interview scheduling, and email notifications. Thus, the system provides a centralized environment, that integrates conventional academic management with learning and placement-related AI-assisted functionalities.

III. PROPOSED SYSTEM

The proposed system focuses on academic, AI-based learning, examination evaluation, campus placement, and student portal. The process consists of conventional role-based management and the utilization of Artificial Intelligence, Large Language Models, Retrieval-Augmented Generation to minimize manual processes and increase the accessibility to educational resources. The proposed system includes user authentication and role-based management, academic records management, uploading and processing of study materials, document retrieval, summary, Multiple Choice Questions (MCQ), examination answer evaluation, placement eligibility verification, interview scheduling, and email alerts. The portal provides various features for administrators, teachers, and students, and the backend manages the application data and AI processes. The complete processing workflow is shown in Fig. 1 below.

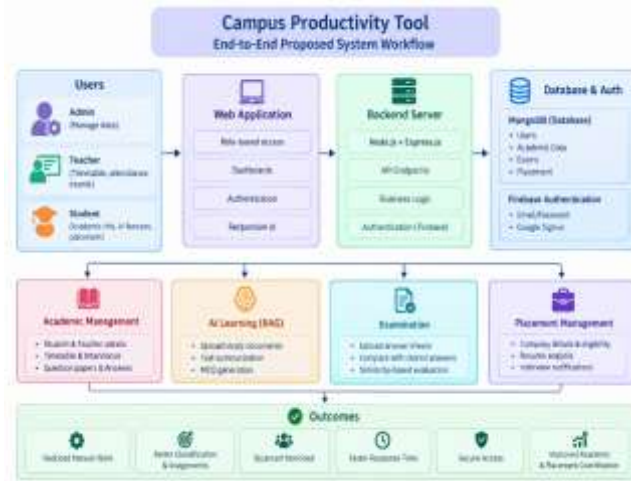


Fig. 1: End-to-End Proposed System Workflow

A. Role-Based User and Academic Management

The proposed system provides a centralized portal for academic management and has three (3) user roles, includes Administrator, Teacher, and Student.

The user authentication and authorization module enables the users to perform various operations on the portal

depending on the assigned roles. For instance, the administrators manage student and teacher data, company data, eligibility criteria, interview schedules, and examination data. The teachers handle time-table, attendance, question papers, and correct answers. The students manage their academic data and upload study materials and resume.

The portal enables role-based management of academic and placement data instead of utilizing multiple independent systems. For instance, student and teacher data, academic records, examination, and placement information are managed within the same portal to increase accessibility and ease of management.

B. AI-Based Document Processing

The proposed system provides an AI module that enables processing of uploaded documents by the students. The uploaded documents undergo pre-processing, including extraction of relevant text, cleaning, and breaking the text into smaller segments or chunks. The processed text is then converted to embeddings using an embedding model.

The obtained embeddings are stored in a vector database for the document retrieval process.

Instead of comparing query embeddings with the document embeddings, the system compares query embeddings with relevant chunk embeddings to ensure that the context of the long study materials is captured.

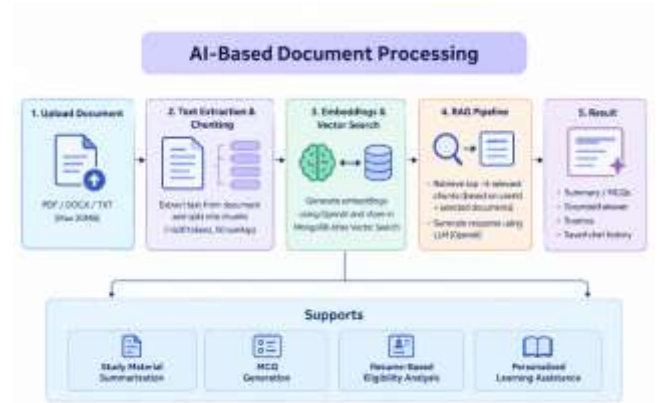


Fig. 2: AI-Based Document Processing

C. Retrieval-Augmented Generation (RAG)

The proposed system utilizes the Retrieval-Augmented Generation (RAG) paradigm to provide document-assisted educational resources. The RAG process works by first comparing the query embeddings (query of the student) with the embeddings of the stored documents to identify relevant chunks.

The identified document chunks and the query are then combined to form the context, which is fed into the Large Language Model to generate the appropriate response. The response then becomes based on the student's query and the relevant portions of the long study materials.

Instead of relying on the model's training data, RAG-based systems utilize external information to generate appropriate responses. Other studies have utilized RAG in education to develop context-aware tutors while recent studies have demonstrated the effectiveness of using RAG in enhancing grounding of educational content generation.



Fig. 3: Retrieval-Augmented Generation Pipeline

The RAG pipeline consists of several modules, including document collection (long study materials uploaded by the student), text extraction, pre-processing, document chunking, and embeddings generation. The embeddings are stored in a vector database. In the generation phase, the system processes the query, compares it with the document embeddings to identify relevant chunks, combine them to form context, and feed it into the LLM. The key components of the RAG pipeline are shown in Fig.3.

D. AI-Based Summary and MCQ Generation

The proposed system can generate meaningful summaries and Multiple Choice Questions (MCQ) from the relevant portions of the uploaded documents. The summary module is capable of capturing the essence of the retrieved context to enable students to quickly understand the important aspects of the long study materials. The MCQ generator can then be utilized to produce multiple-choice questions from the retrieved context. The generated MCQs contain appropriate options and answers based on the context. The modules can then assist the students in fully comprehending the long study materials while minimizing the effort required to extract key points and generate questions. Several studies have demonstrated the effectiveness of utilizing Large Language Models and RAG-based context retrieval to generate summaries and MCQs.

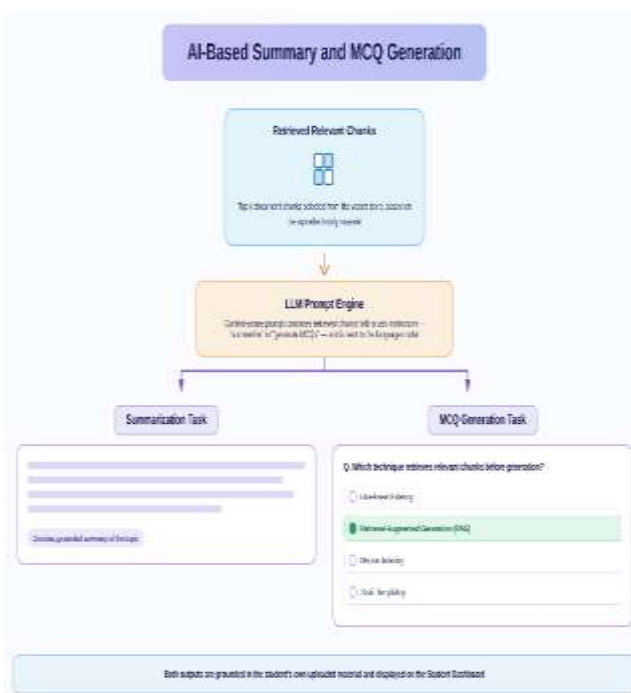


Fig. 4: AI-Based Summary and MCQ Generation

The summary and MCQ generation process involves uploading the long document, extracting relevant text, chunking the text, generating embeddings, retrieving the relevant text portions, and passing them to the LLM to generate the summary and questions. The overall workflow is shown in Fig.4.

E. Examination and Answer Evaluation

The proposed system can be utilized to manage examinations and evaluate answers. For instance, the teachers can upload or store the question papers and correct answers while the students' answers can be compared to the correct answers using the similarity module.

The similarity scores indicate how close the students' answers are to the correct answers. The comparison minimizes the amount of manual effort required to evaluate the students' answer scripts. The project report indicates that the accuracy can be achieved at around 80-90% depending on the similarity scores and the quality of the documents.



The examination evaluation process consists of the question paper or correct answer, student's answer, text processing, and similarity calculation. The complete workflow is demonstrated in Fig.5.

F. Campus Placement and Eligibility Management

The proposed system can be utilized to manage the campus placements, including the company information, eligibility criteria, and interview schedules. The system consists of a campus placement module that manages the company data and eligibility requirements. The administrators can then manage the company data and set the eligibility criteria. The student's academic data can then be compared to the eligibility criteria to determine whether they qualify for the placement or not. The comparison minimizes the amount of manual effort required to evaluate the students' eligibility based on their academic data.

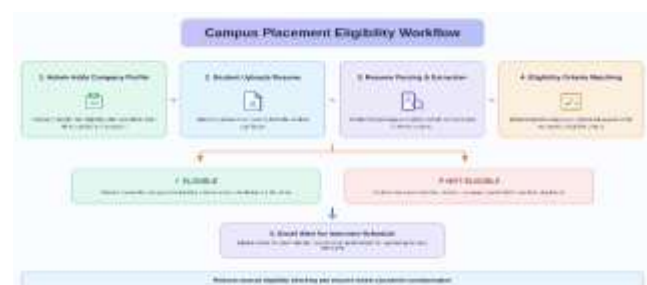


Fig. 6: Campus Placement Eligibility Workflow

The placement eligibility workflow consists of the company profile, eligibility criteria, student academic data, comparison, and interview schedule. The eligibility criteria are compared to the student academic data to determine whether they qualify for the placement. The complete process is shown in Fig.6.

G. Interview Scheduling and Email Notifications

The proposed system can be utilized to schedule interviews and provide email notifications to the students concerning their interview schedules. The administrators can manage the interview data in addition to linking the information to the appropriate company and placement data. Email alerts can also be utilized to notify the students of any placement-related information or updates.

This feature ensures that the placement-related information is centralized instead of utilizing multiple communication channels to deliver the interview schedules. The interview scheduling information can be stored together with the placement data and company details.

H. Integrated System Architecture

The complete proposed system consists of the role-based management, document processing, RAG-based summary and MCQ generation, examination processing, and campus placement modules. The administrator, teacher, and student roles can be utilized to operate on the student portal while the backend manages the data in the database. The backend oversees the academic, examination, and placement data while the AI document processing module manages the study material data.

The RAG-based context retrieval enables the system to generate summaries and MCQs based on the uploaded documents. The examination module compares the students' answers to the correct answers using the similarity metric. The placement eligibility module compares the student's academic data to the specified requirements. The overall workflow can then be summarized using the diagram below.



Fig. 7: Integrated System Architecture

The proposed system is a centralized student portal that provides academic management, AI-assisted learning,

examination evaluation, and campus placement. The document-based processing module increases the accessibility to long study materials by enabling the system to fetch relevant portions of the text for summary generation and MCQ creation. The examination evaluation relies on the similarity metric to minimize the amount of manual effort spent on marking the answer scripts.

The campus placement enables the system to compare the student's academic data to the specified eligibility criteria to minimize the manual work required in assessing the placement eligibility. Unlike the conventional approaches that only utilizes certain aspects of chatbot, document processing, examination marking, and placement, the proposed approach integrates these processes into one system that utilizes the RAG paradigm to increase accessibility to educational resources. Several recent studies highlight the effectiveness of employing RAG and Large Language Models in education to provide question-answering, personalized tutoring, and automated marking systems.

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION

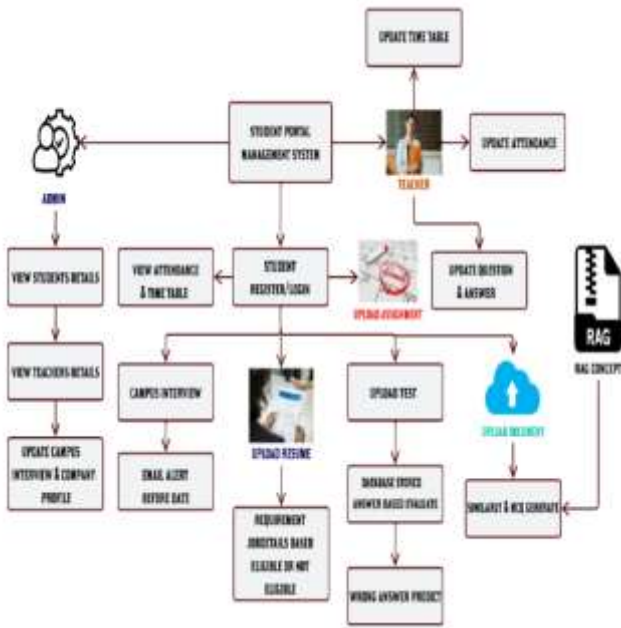
The proposed Campus Productivity Tool is a role-based centralized application that addresses academic management, document summarization, multiple choice question (MCQ) generation, and campus placement management. The application offers distinct features for administrators, teachers, and students, while maintaining a unified approach to academic and placement management. The uploaded study materials are processed through a Retrieval-Augmented Generation (RAG) pipeline for implementing document summarization and MCQ generation, and the examination module performs similarity-based evaluation of the answers provided by students. This section discusses the system architecture, technologies, database design, AI document processing, and access control of the proposed application.

A. System Architecture

The proposed campus productivity tool is designed according to the modular architecture that accommodates the web application interface, application/backend services, database, document processing module, vector retrieval, and Large Language Model (LLM). Student, teacher, and administrator roles access the system through the unified web application interface.

The study materials uploaded by students are processed through the document processing pipeline that performs text extraction, preprocessing, chunking, and embedding. The resultant embeddings are stored in the vector database, which can be queried according to the similarity scores between the input query and contexts in the stored documents. The retrieved information is used as input to the LLM to generate document summaries and MCQs. RAG-based system architectures have been widely considered and implemented for personalized learning and question-answering applications.

The examination module compares student-provided answers to the pre-specified correct answers using similarity-based scoring. The placement module processes company eligibility requirements and interview schedule



The overall architecture of the system can be summarized as:

Administrator/Teacher/Student → Web Application →
Backend Services → Database & Document Storage →
AI/RAG Processing →
Summary/MCQ/Evaluation/Placement Results

B. Technologies

The proposed campus productivity tool utilizes web technologies, database systems, artificial intelligence (AI), and natural language processing (NLP) technologies and techniques. The implemented technologies and components in the proposed system are illustrated in Fig. 7.

The RAG pipeline that underpins the system's document summarization and MCQ generation follows the standard retrieval and generation paradigm wherein information is retrieved from an external knowledge source and used to fine-tune the language model for specific tasks.

Educational researchers have increasingly adopted RAG-based approaches to develop personalized learning, tutoring, and question-answering applications.



Fig. 7. Technologies used in the proposed system.

C. Database Design

The database accommodates the information managed by

the proposed campus productivity tool, which includes user profiles, roles, academic records, attendance/timetable, examinations, company profiles, eligibility requirements, interview schedules, and placement records.

The database design follows referential integrity to ensure that relevant information is accessed and updated accordingly across the modules.

The general design involves establishing foreign key constraints to ensure that user-related information corresponds to the user roles and student academic records correspond to the students' details. Similarly, examination-related information is linked to the question-answer details, while company-related information is associated with eligibility requirements and interview schedules.

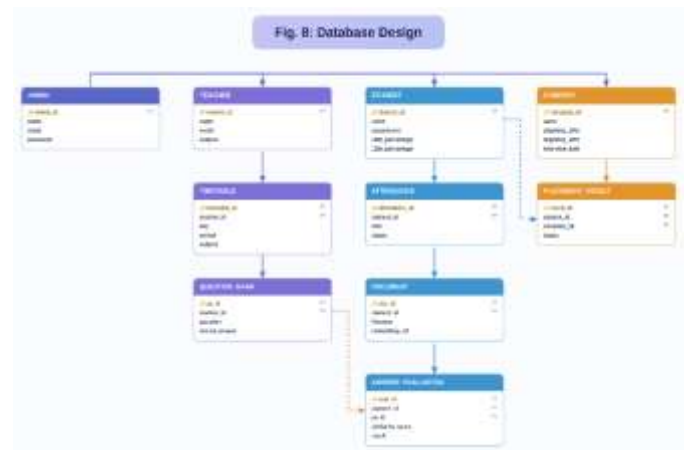


Fig. 8. Database design

The entities in the database design include:

Users – stores student, teacher, and administrator information.

Roles – stores permission levels and features accessible by the three user roles.

Students – stores student-related academic information.

Teachers – stores teacher-related information.

Timetable – stores academic timetable-related information.

Attendance – stores attendance-related information.

Study Materials – stores information about the study materials uploaded by students for AI processing.

Documents/Chunks – stores processed document information used in the RAG pipeline.

Examinations – stores examination-related information.

Answers – stores correct and student-provided answers used in the examination module.

Companies – stores information about companies involved in campus placements.

Eligibility Criteria – stores eligibility requirements specified by companies.

Interview Schedules – stores interview-related information.

The foreign key constraints and relationships among these entities ensure consistent access and update of student, academic, examination, and placement records in the

proposed system.

D. RAG Knowledge Base and Vector Database

The proposed campus productivity tool implements a RAG pipeline to process the documents uploaded by students for generating document summaries and MCQs. The document is processed through a series of steps, including text extraction, preprocessing, chunking, and embedding, to generate document embeddings that are stored in the vector database.

Upon receiving a query or a prompt, the system generates an embedding of the query and compares it with the stored document embeddings to identify similar contexts and retrieve the relevant information chunks. These chunks are then used as input to the large language model to generate document summaries or MCQs.



Fig. 9. RAG pipeline

The complete document retrieval process involves:

Study Material → Text Extraction → Preprocessing → Chunking → Embedding Generation → Vector Database → Query Embedding → Similarity Search → Relevant Chunks → LLM

The RAG pipeline enables the system to offer context-based information retrieval and generation of summaries and MCQs in natural language processing (NLP) applications. Researchers have increasingly adopted RAG systems to enhance the performance of conventional question-answering systems, particularly in addressing the limitations of the unsupported generation of responses.

E. AI-Based Summary and MCQ Generation

The retrieved document chunks are used to generate two types of learning contents, namely, summaries and multiple choice questions (MCQs).

For summary generation, the retrieved context is provided as input to the large language model (LLM) to generate a succinct summary of the document contents. For MCQ generation, the retrieved context is used to generate multiple choice questions, including the question prompt and answer options, alongside the correct response.



Fig. 10. AI-based summary and MCQ generation

The document context can be summarized and transformed into multiple-choice questions as:

Retrieved Document Context → LLM → Summary / MCQs → Student

The use of RAG in conjunction with large language models (LLMs) for MCQ generation has been widely explored in recent research works. For instance, the MCQGen system utilizes retrieval-augmented generation and prompt engineering to implement AI-based MCQ generation for educational purposes.

F. Examination Answer Evaluation

The examination module enables similarity-based comparison of student-provided answers to the pre-specified correct responses.

Teachers can add question papers and the corresponding correct answers using the examination module. Furthermore, the evaluation component scores the student responses based on the similarity information between the student response and the correct answer, thus reducing the overall effort entailed in the manual marking of examinations.

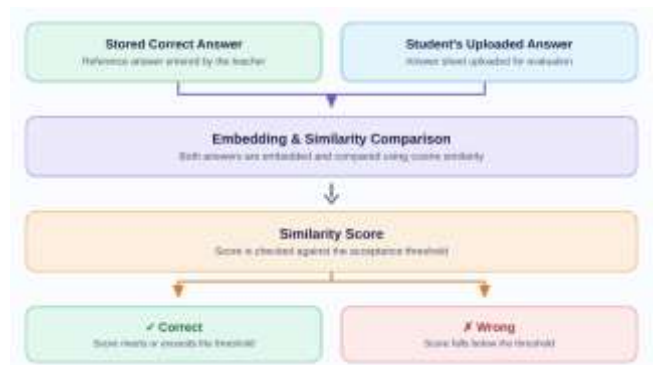


Fig. 11. Examination answer evaluation

The similarity scores can be determined using:

Question / Correct Answer → Student Response → Text Processing → Similarity Calculation → Evaluation Result

The report states that the expected performance of the examination answer evaluation component is between 80% and 90%, depending on the similarity scores and text processing.

G. Campus Placement and Eligibility Management

The campus placement component enables the management of company profiles, eligibility criteria, student academic information, and interview schedules. The eligibility requirements specified by companies can be compared against the student academic information to

determine eligibility. The following workflow represents the general process of determining student eligibility:

Company Profile → Eligibility Criteria → Student Academic Data → Criteria Comparison → Eligibility Result → Interview Schedule

The campus placement module therefore offers a centralized platform for managing placement-related information, including company profiles, eligibility requirements, and interview schedules. The placement-related information can be stored in the system, and email notifications sent to the relevant students based on the eligibility and interview schedules.

H. Role-Based Access Control

The proposed campus productivity tool implements role-based access control to ensure that only the authorized roles access specific features and information. The system accommodates three major roles, namely, Administrator, Teacher, and Student.

The administrator role has access to all the modules, including user management, placement management, and examination management. The teacher role has access to the timetable, attendance, question papers, and answer management modules. The student role can view the academic information, upload study materials for AI processing, generate summaries and MCQs, submit placement-related documents for eligibility determination, and view relevant placement eligibility and interview schedules.

The following workflow illustrates the role-based access:

Administrator → User Management + Placement Management + Examination Management

Teacher → Timetable + Attendance + Question Papers + Answer Management

Student → Academic Information + Study Materials + AI Processing + Placement Information

The role-based access ensures that the academic and placement-related information is appropriately managed while maintaining the overall centralization of the system.

I. Integrated Implementation Workflow

The proposed campus productivity tool entails an integrated implementation workflow that embodies the discussed components and features in a unified manner. Users first log in and gain access to the role-specific dashboard.

The academic and placement-related information is stored in the database, whereas the uploaded study materials undergo AI processing to generate document summaries and MCQs. The examination answers are scored using the similarity-based evaluation procedures, and the placement eligibility and interview schedules are determined by comparing the student academic information against the eligibility criteria.

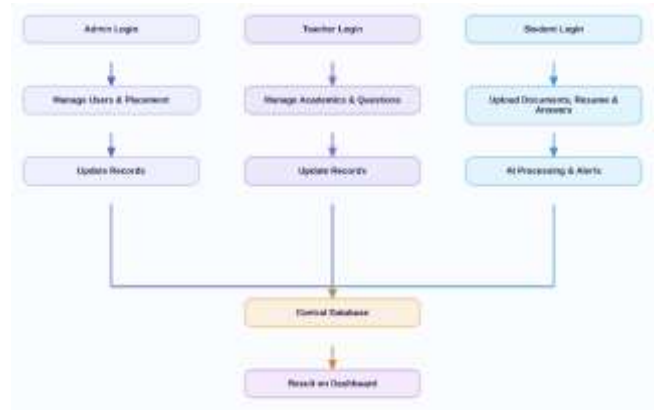


Fig. 12. Complete system implementation workflow

The overall system implementation workflow is summarized as:

User Authentication → Role-Based Dashboard Access → Academic / AI / Examination / Placement Module Access → Backend Processing → Database / Vector Database → Results and Notifications

The proposed campus productivity tool therefore combines conventional student management features with AI-assisted academic processing and placement management in a single application. The overall architecture is also consistent with recent RAG-based educational applications that incorporate separate indexing, retrieval, generation, and user-interface modules.

V. EXPERIMENTAL SETUP

A. Dataset and Document

The proposed system is not based on a fixed set of classified data set but on student-uploaded material. The students may upload their notes and course material along with relevant documents and related information. The uploaded document is used by the AI-based document processing pipeline to generate summaries and multiple-choice questions.

The uploaded material is aggregated and preprocessed prior to feeding into the RAG pipeline. Since the proposed system is student-specific, the knowledge base is generated from the uploaded documents for contextualized retrievals. This implies that the students' own learning materials are mined to generate AI-based content.

The experimental input falls under the following categories:

The use of document-based input is common in the RAG-based educational AI space where external content is retrieved and fed into an LLM for context-specific tasks.

B. Document Preprocessing

Prior to retrieval, the uploaded student material is preprocessed to extract relevant information. The first step involves extracting text from documents followed by cleaning and chunking to generate smaller document chunks. These chunks are embedded to generate vector representations and stored in a vector database.

The overall document preprocessing pipeline takes the form of:

Uploaded Document → Text Extraction → Cleaning → Chunking → Embedding → Vector Database



Fig. 10. Document Preprocessing and Embedding

The report mentions LangChain, LlamaIndex, FAISS, ChromaDB, Pinecone, and Weaviate as potential technologies for implementing the RAG-based document retrieval pipeline.

C. RAG-Based Experimental Setup

The main AI experiment centers around the Retrieval-Augmented Generation paradigm. Rather than feeding the entire document to the language model, only the relevant portions are retrieved and are used in conjunction with the query. This is achieved by first embedding the query and comparing it to the document embeddings to find relevant chunks.

Once the most relevant document chunks have been retrieved, they are combined with the query to form the context for the LLM. The overall RAG pipeline takes the form of:

Student Query → Query Embedding → Similarity Search → Relevant Document Chunks → Context Formation → LLM → Output

The output from the RAG pipeline consists of document summaries and multiple-choice questions. RAG-based educational AI systems have been used in recent studies for context-based learning and MCQ generation.

D. Summary Generation Experiment

The summary experiment is aimed at testing the ability of the proposed system to generate abridged versions of long documents. For each document, relevant chunks are retrieved based on the student query or requested content. The retrieved chunks are then used as context for the LLM to generate a summary. This will help the student in accessing important information without having to go through the entire lengthy document.

The overall pipeline for the summary generation experiment is as follows:

Study Material → Preprocessing → Chunking → Embedding → Retrieval → Context → LLM → Summary

The generated summary is likely to contain the salient concepts from the original document while being shorter and more concise.

E. MCQ Generation Experiment

The second major AI experiment centers around

generating Multiple Choice Questions (MCQ) from the uploaded study material. Similar to the summary generation experiment, relevant document chunks are retrieved and are used for generating MCQ-based questions. The generated MCQ can then be used for self-assessment by the students.

The overall pipeline for the MCQ generation experiment is as follows:

Study Material → RAG Retrieval → Relevant Content → LLM → Question Generation → Options → Correct Answer

There have been several recent works, including the MCQGen system, which use an LLM in conjunction with a RAG pipeline for automatic MCQ generation from external content.

F. Examination Answer Evaluation

The examination experiment revolves around comparing the student answer to a reference answer. The model accepts the question, reference answer, and student answer and uses a similarity metric to determine the accuracy of the student answer.

The evaluation procedure can be summarized as follows:

Correct Answer + Student Answer → Text Processing → Similarity Metric → Evaluation

Ideally, the similarity between the reference and student answer determines the evaluation outcome. According to the project report, the expected accuracy for the answer evaluation component is around 80–90% depending on the answer content and similarity.

G. Placement Eligibility Experiment

Unlike the other experiments, the placement component does not require a training set. Rather, student academic information is compared to company-specific eligibility criteria to determine suitability. For each company, the placement component stores the eligibility requirements, which are then compared to the student's academic information.

The overall pipeline for the placement experiment is summarized as follows:

Company Eligibility → Student Academic Info → Comparison → Eligibility Outcome

The placement experiment, therefore, is evaluating whether the system can appropriately compare student information to eligibility requirements and connect the student to the relevant placement/interview details.

I. Evaluation Criteria

The evaluation criteria for the proposed system depend on the component under consideration. For the RAG pipeline, one can use relevance and contextual accuracy as evaluation criteria. This would entail analyzing whether the retrieved document chunks contain relevant information for the given query. For summary generation, evaluation criteria include relevance, completeness, and consistency with the source material. For the MCQ generation experiment, one can employ correctness, relevance, answerability, and alignment with the uploaded content. Automatic evaluation methodologies have been used in several recent studies for RAG-based educational AI and MCQ generation systems.

For the examination component, the evaluation metric involves comparing the student answer to a reference answer. On the other hand, the placement component can be evaluated by comparing the system outcomes to the expected eligibility outcome for a given student.

VI. RESULTS AND DISCUSSION

The proposed Campus Productivity Tool integrates role-based academic management, AI-based document processing, Retrieval-Augmented Generation (RAG), examination answer evaluation and campus placement management features into a single system. It was designed to reduce manual effort in academic activities and provide students with AI-assisted access to their study material. This section presents the implemented system interface, document processing workflow, RAG-based content generation, examination evaluation and placement management functionality.

A. User Interface

The proposed system offers a role-based interface for Administrators, Teachers and Students, with each role being provided with functionalities according to its roles within the academic and placement environment.

The Administrator interface provides facilities for managing student and teacher profiles, company information, eligibility requirements, interview schedules and other administrative activities. The Teacher interface supports timetable management, attendance management, question paper management and correct-answer management. The Student interface provides access to academic information and allows students to upload study materials and resumes for the corresponding AI and placement functionalities.

The centralized interface allows users to access multiple academic and career-related functions without depending on separate applications.



Fig. 12. Role-Based User Interface

B. AI-Based Document Processing Results

The document-processing module was implemented to allow students to upload study materials for AI-assisted processing. After a document is uploaded, the system extracts the available text and preprocesses it before dividing the content into smaller chunks.

The processed chunks were then converted into vector representations and stored for semantic retrieval. When a

student submits a query, the system retrieves the most relevant sections of the uploaded document and uses them as contextual information for subsequent generation.

The results demonstrate that the document-processing pipeline can transform uploaded academic materials into a searchable knowledge source for the RAG module.

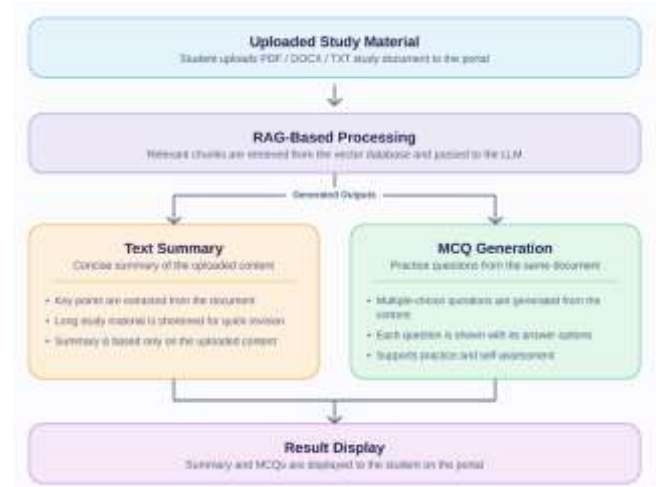


Fig. 13. Uploaded Study Material and Processing Result

C. RAG-Based Summary Generation

The RAG module was used to generate summaries from student-uploaded study materials. Instead of passing the complete document directly to the language model, the system first retrieves relevant content based on semantic similarity.

The retrieved content is then provided to the language model as context, allowing the generated summary to remain associated with the source material. This is particularly useful when students upload lengthy academic documents and require a concise representation of the important concepts.

The summary-generation workflow is:

Uploaded Study Material → Text Processing → Chunking → Embedding → Semantic Retrieval → Relevant Context → LLM → Generated Summary

The resulting summaries offer students a more concise representation of their study materials. Recent research also reports the usefulness of RAG for grounding educational content in source material and improving the reliability of generated learning content.

D. MCQ Generation Results

The proposed system also generates Multiple-Choice Questions (MCQs) from uploaded educational materials. Relevant document content is retrieved through the RAG pipeline and supplied to the language model as context.

The generated questions contain the question statement, answer options and corresponding correct answer. Since the generation is based on retrieved study material, the questions are intended to remain relevant to the content provided by the student.

The MCQ generation workflow is:

Study Material → RAG Retrieval → Relevant Content → LLM → MCQ Generation → Answer

This functionality reduces the manual effort required to prepare practice questions and provides students with an additional mechanism for self-assessment. Existing research has also demonstrated the use of LLMs and RAG for automated MCQ generation in personalized learning environments.

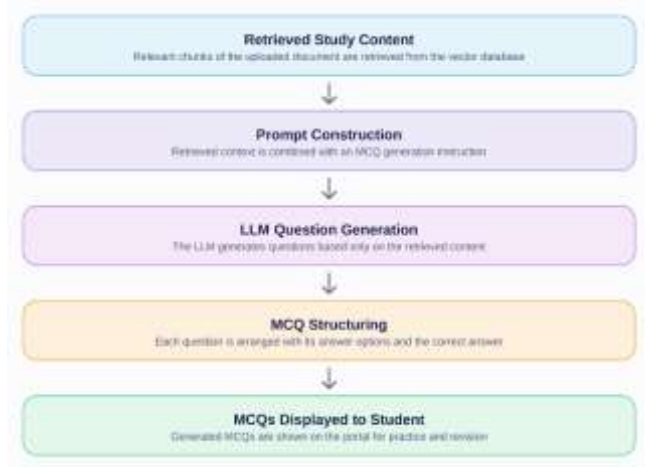


Fig. 14. Generated MCQs from Study Material

E. Examination Answer Evaluation

The examination module provides an automated approach for assisting in the evaluation of student answers. The teacher provides the question and corresponding correct answer, while the student's response is processed by the evaluation module.

The system compares the student answer with the predefined correct answer using a similarity-based approach. The resulting similarity information is used to determine the correspondence between the two responses.

The evaluation workflow is:

Question → Correct Answer → Student Answer → Text Processing → Similarity Calculation → Evaluation Result

The project report specifies an expected effectiveness of approximately 80–90%, depending on the quality of the input and the similarity between the student response and the expected answer. This result should be treated as the project's expected/target effectiveness, unless a measured test-set evaluation has been performed.



Fig. 15. Answer Evaluation Result

F. Campus Placement Results

The campus placement module provides a centralized mechanism for managing company profiles, eligibility criteria, student academic information and interview schedules.

The system compares the academic information of students against the eligibility criteria specified for a company. The resulting eligibility status can then be used to identify students who satisfy the requirements for a particular placement opportunity.

The placement workflow is:

Company Profile → Eligibility Criteria → Student Academic Data → Criteria Comparison → Eligibility Result

The module also supports interview schedule management and email-based notifications, offering students access to relevant placement information through the same platform.



Fig. 16. Placement Eligibility and Interview Schedule

H. Discussion

The implemented system demonstrates the integration of conventional student-management functionality with AI-assisted educational processing. The RAG pipeline enables the uploaded study materials to be converted into a searchable knowledge source, from which relevant content can be retrieved for summary and MCQ generation. This provides a document-grounded approach to educational content generation rather than relying solely on the general knowledge of the language model.

The examination module further reduces manual effort by applying similarity-based comparison between student answers and predefined answers. Similarly, the placement module provides structured eligibility checking by comparing student academic information against company requirements.

The results also indicate that the proposed system is not limited to a single AI function. Instead, it combines academic management, document processing, RAG-based learning assistance, automated answer evaluation and placement management features in one platform. This integrated approach is consistent with recent educational AI research, where RAG is increasingly used alongside LLMs for context-aware question answering, personalized learning and automated educational content generation.

However, the current implementation should be further

evaluated using a larger collection of real student documents and formally annotated test cases. In particular, future evaluation can measure retrieval accuracy, summary quality, MCQ correctness and relevance, answer-evaluation accuracy and placement-eligibility accuracy using quantitative metrics and human evaluation. Recent RAG research emphasizes the importance of systematic evaluation since generated content can vary with document quality, retrieval performance and model configuration.

VII. CONCLUSION

The proposed Campus Productivity Tool aims to provide an integrated platform for managing academic activities, examination process, AI based learning, and campus placement activities to achieve a common web-based unified system. The proposed system utilizes the role-based access for Admin, Teacher and Student, centralized student and teacher management including the timetable, attendance, examination and question paper management, AI based document processing, resume based company eligibility analysis, and campus interview notification. The retrieval-augmented generation (RAG) helps the students to process their uploaded study materials to fetch the relevant summary and multiple-choice questions based on the document content.

The proposed system also assists the examination module by providing the automated answer-sheet evaluation by comparing the student's response with the stored answers. Placement module helps the students to find the relevant companies based on their eligibility and notify the upcoming campus interview dates.

The proposed Campus Productivity Tool is expected to minimize the manual efforts in academic management, provide access to relevant academic information, and enable personalized learning, examination evaluation and campus interview organization to improve the overall productivity.

VIII. FUTURE WORK

The future scope of work considers the improvement of the existing capabilities, accuracy, scalability, and AI based features of the Campus Productivity Tool. The AI based document management module can be improved by allowing the support for additional document formats and optimizing the RAG pipeline for better performance. The mechanisms for providing summary and MCQs, along with the answer-sheet evaluation can be improved for providing the personalized learning experience based on the student's performance and learning needs.

The answer-sheet evaluation module can be improved by utilizing the semantic similarity models to achieve higher accuracy for different answer formats and document quality. Future development can also consider the addition of new features related to placements such as application tracking, personalized placement recommendations, interview preparation, and resume analysis.

The notification system can also be expanded by providing the additional channels and advanced reminder mechanisms. The system can also be enriched with additional dashboards for students, teachers and admin, language localizations, security improvements, access

control improvements, and system integration to provide a better unified experience. All these future development opportunities will enable the transformation of the Campus Productivity Tool into a comprehensive and scalable platform that effectively empowers the students in their academic learning and campus placement activities.

IX. REFERENCE

- [1] T. Nguyen, Q. Pham, L. Chen, and R. Zhang, "A review on personalized learning and intelligent tutoring systems using deep learning and graph-based approaches," *IEEE Access*, 2023. [Online] Available: <https://ieeexplore.ieee.org/abstract/document/11386890/>
- [2] A. A. Syed, F. L. Gaol, and T. Matsuo, "A survey of the state-of-the-art models in neural abstractive text summarization," *IEEE Access*, vol. 9, pp. 13248–13265, 2021, doi:10.1109/ACCESS.2021.3052783.
- [3] A. Elsaid, A. Mohammed, L. F. Ibrahim, and M. M. Sakre, "A comprehensive review of Arabic text summarization," *IEEE Access*, vol. 10, pp. 38012–38030, 2022, doi:10.1109/ACCESS.2022.3163292.
- [4] S. Tanwar, Q. Bhatia, P. Patel, A. Kumari, P. K. Singh, and W.-C. Hong, "Machine learning adoption in blockchain-based smart applications: The challenges, and a way forward," *IEEE Access*, vol. 8, pp. 474–488, 2020, doi: 10.1109/ACCESS.2019.2961372.
- [5] B. Khan, R. Naseem, F. Muhammad, G. Abbas, and S. Kim, "An empirical evaluation of machine learning techniques for chronic kidney disease prophecy," *IEEE Access*, vol. 8, pp. 55012–55022, 2020, doi: 10.1109/ACCESS.2020.2981689.
- [6] C. W. Tan, L. Ling, P.-D. Yu, C. N. Hang, and M. F. Wong, "Mathematics gamification in mobile app software for personalized learning at scale," in *Proc. IEEE Integr. STEM Educ. Conf. (ISEC)*, Aug. 2020, pp. 1–5.
- [7] D. R. Ch and S. K. Saha, "Automatic multiple choice question generation from text: A survey," *IEEE Trans. Learn. Technol.*, vol. 13, no. 1, pp. 14–25, Jan. 2020.
- [8] Y. Chen, W. Huang, and K.-S. Lee, "Modeling knowledge acquisition with graph neural networks in educational systems," *IEEE Trans. Learn. Technol.*, 2024. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/11386890/>
- [9] F. K. Sufi and I. Khalil, "Automated disaster monitoring from social media posts using AI-based location intelligence and sentiment analysis," *IEEE Trans. Computat. Social Syst.*, early access, Mar. 18, 2022, doi: 10.1109/TCSS.2022.3157142.
- [10] R. Pramana, Debora, J. J. Subroto, A. A. S. Gunawan, and Anderies, "Systematic literature review of stemming and lemmatization performance for sentence similarity," in *Proc. IEEE 7th Int. Conf. Inf. Technol. Digit. Appl. (ICITDA)*, Nov. 2022, pp. 1–6.

